Ananya Uppal

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Summary

Master of Computer and Information Technology grad student (GPA 4.0) at Purdue University with experience in translating ambiguous requirements into production-ready systems and communicating complex findings to technical and non-technical stakeholders. Built evaluation frameworks and API-integrated data pipelines at Palo Alto Networks and Leen. Research background in ML pipeline fairness, calibration, and label quality.

Education

Purdue University – MS in Computer and Information Technology **GPA: 4.0/4.0** **Aug 2024 – June 2026**
Coursework: Responsible Data Science, AI, Statistics, HCI. ML data pipeline research with Dr. Romila Pradhan, RDS Lab. Graduate teaching assistant across 4 semesters; Taught Data Management (CNIT 392) and Visual Programming (CNIT 175).

PES University – B.Tech in Computer Science **GPA: 9.06/10** **Aug 2019 – May 2023**
Coursework: Big Data, NLP, IR, Network Analysis, DBMS, OOD. 6-time MRD Scholarship winner (top 20% of class).
Club Member of Association for Computing Machinery-Women and The Changemaker's Society: Organised and mentored Hackathons; Member of career panels to mentor students with job hunting.

Skills

Languages & Databases	Python, C/C++, Git, Ruby, SQL, Java, MATLAB, SAS, MongoDB, React.js, HTML
Technical Tools	Docker, REST APIs, Git, Hadoop, Spark, Linux, Jenkins, LLMs, GenAI, AWS, GCP
Skills	Responsible ML/AI, ML pipeline engineering, Prompt Engineering, API Dev, Test automation

Experience

Data Science Intern, **Leen**, *San Francisco, CA* **Jul 2025 – Aug 2025**
- Designed 3 new connectors for API-based data models to map and consolidate GRC entity, evidence, assessment, and controls data from ServiceNow GRC, OneTrust, and Archer.
- Drove requirements discovery meetings with customers, translated business needs into data models and API connector specs, and led deployment to production, directly contributing to revenue.
- Successfully signed on and onboarded new customers in the span of a few weeks with significant revenue contribution in a startup environment.

Software Engineer, **Palo Alto Networks**, *Bengaluru, KA* **Aug 2023 – Jul 2024**
- Designed and developed QA and evaluation frameworks for firewall software across VM and hardware device families (3k/5k/7k series); defined structured pass/fail rubrics and coverage thresholds, improving code coverage from 23% to 78%.
- Built Jenkins-based testing pipelines and communicated evaluation findings and reliability metrics to cross-functional engineering and product teams.
- Validated and automated tests for root certificate integrity at scale across 8 PanOS releases and 12M+ devices; developed systematic sanity quality test suites by running suites against internal git PR's.

Software Engineer Intern, **Palo Alto Networks**, *Bengaluru, KA* **Jan 2023 – Jul 2023**
- Reduced automation backlog 30% by designing and integrating regression test suites (pytest, gnmic); expanded REST API coverage across six PanOS releases using Selenium for streaming telemetry validation.

Software Engineer Intern, **Nasdaq**, *Bengaluru, KA* **Jun 2022 – Jul 2022**
- Evaluated 4 blockchain platforms against 17 technical criteria using Hyperledger Caliper; synthesized quantitative findings into clear recommendations presented to non-technical and executive stakeholders.

Projects & Research

Data Valuation for Label Error Detection, *MS Thesis, Purdue RDS Lab* **Aug 2024 – Present**
- Evaluating and comparing Shapley-value-based and entropy based methods to detect and repair mislabeled/noisy label data in ML pipelines; focus on improving model fairness(removal of bias), reliability, and explainability.
- Designed experimental evaluation framework to measure detection accuracy and fairness impact across multiple datasets.

Chemical Identification Tool, *Dow Chemical, TDM 511* **Jan 2025 - May 2025**
- Translated ambiguous domain requirements from a chemical engineering customer into a structured ML specification; Worked with corporate mentors to develop a multi-stage XGBoost model predicting chemical peroxide formation using feature engineering (ionic charges, molecular weights) and systematic feature selection.

Interpretable Hybrid Recommender via Graph Convolution(CCCE'23, Stockholm)
- Trained a 4-model hybrid model utilising GCNs to augment recommendation serendipity. Formulated a novel distance-based serendipity metric and improved interpretability through KNN feature importance analysis

RePI: Research Paper Impact Analysis (ISDA'22)